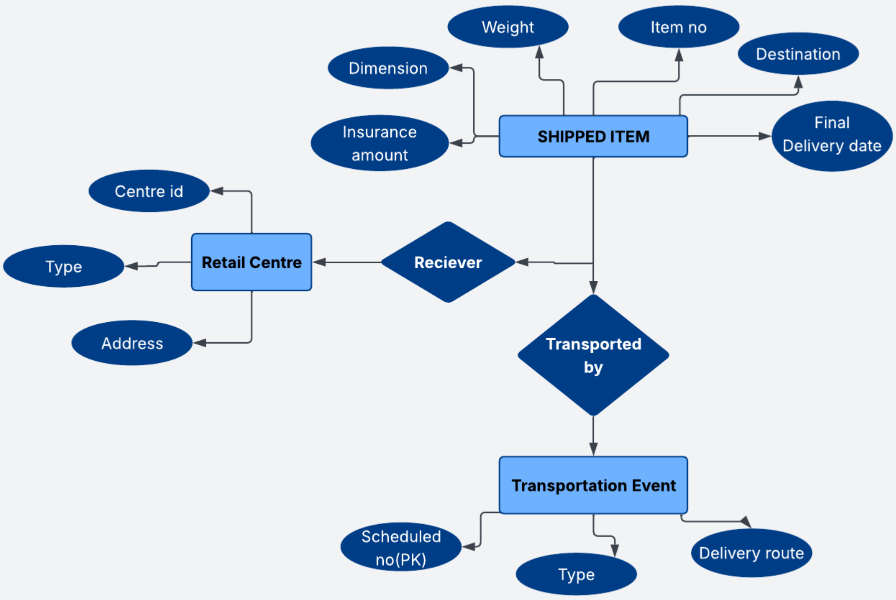
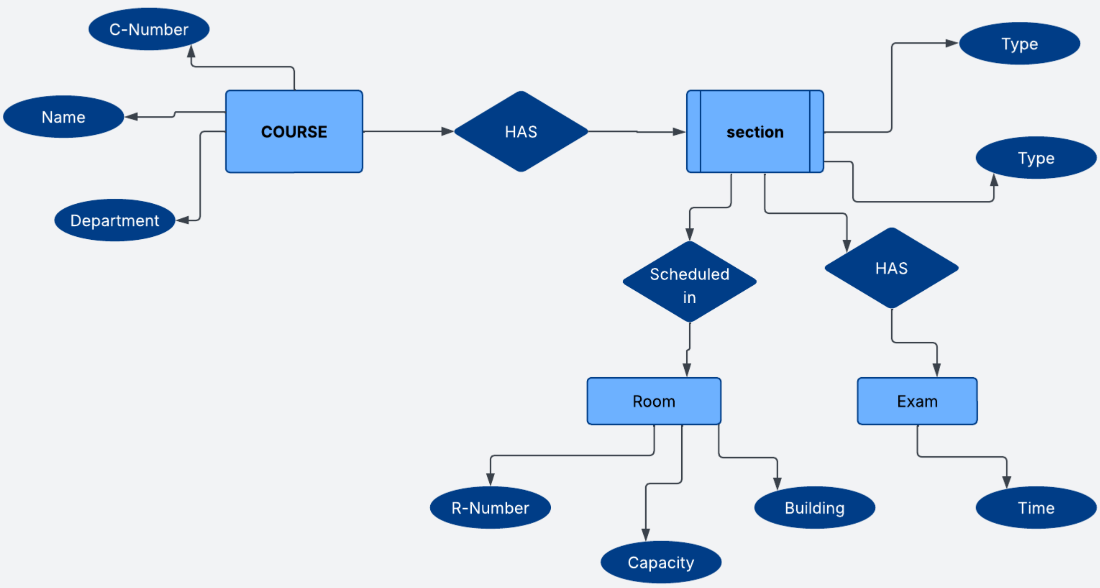
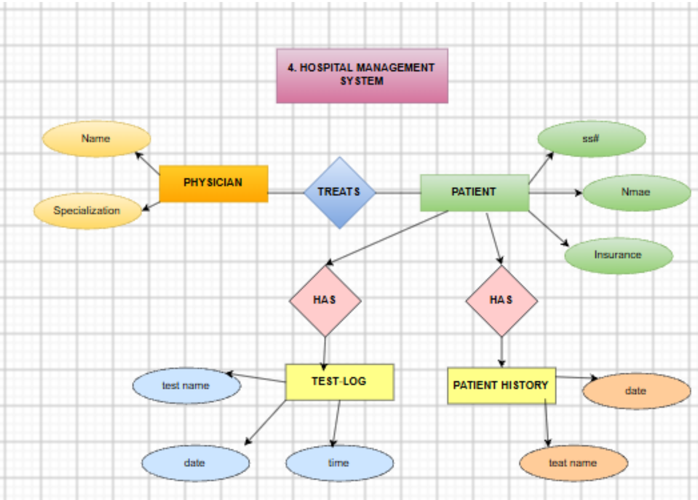
1.UPS PRODUCT TRACKING INFORMATION SYSTEM

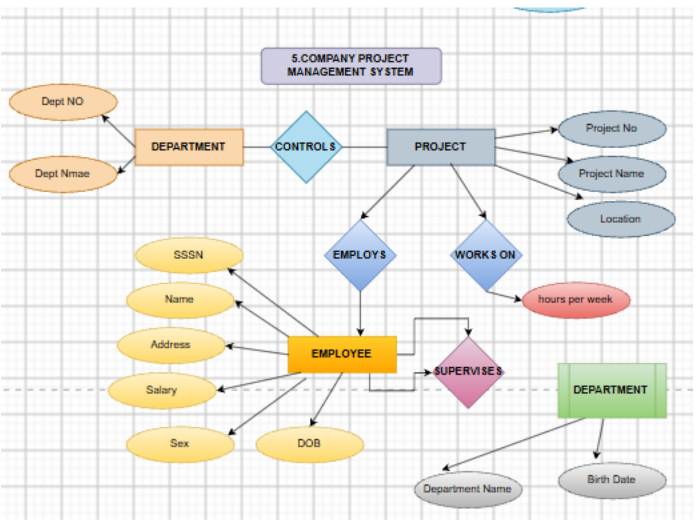


2.COURSE SYSTEM MANAGEMENT




3.UNIVERSITY REGISTER'S OFFICE
Course no
title
credits
syllabus
prerequisites
COURSE
HAS
year
semester
section
timings
COURSE OFFERING
grade
TEACHES
ENROLLS
internship id
name
INSTRUCTOR
STUDENT
studentid
nasme
program


4. HOSPITAL MANAGEMENT SYSTEM
Name
Specialization
PHYSICIAN
TREATS
PATIENT
ssn
Nmae
Insurance
HAS
HAS
test name
TEST-LOG
date
time
PATIENT HISTORY
date
teat name


5.COMPANY PROJECT MANAGEMENT SYSTEM
Dept NO
Dept Nmae
DEPARTMENT
CONTROLS
PROJECT
Project No
Project Name
Location
EMPLOYS
WORKS ON
hours per week
SSSN
Name
Address
Salary
Sex
DOB
EMPLOYEE
SUPERVISES
DEPARTMENT
Department Name
Birth Date

6. Schema and Instance

Schema

A **schema** is the blueprint or structure of a database. It defines the tables, attributes, relationships, and constraints. It changes very rarely.

Example:

Student(Roll_No, Name, Department, Marks)

Instance

An **instance** is the actual data stored in the database at a particular time. It changes whenever data is inserted, updated, or deleted.

Example:

Roll_No	Name	Department	Marks
101	Rahul	CSE	90
102	Priya	ECE	85

Types of Schema

Logical Schema

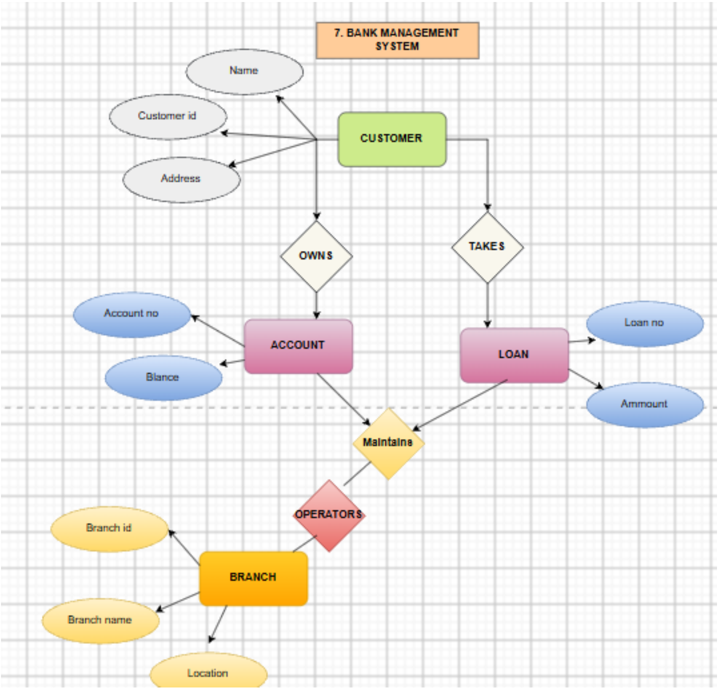
- Describes the logical structure of the database.
- Defines tables and relationships.

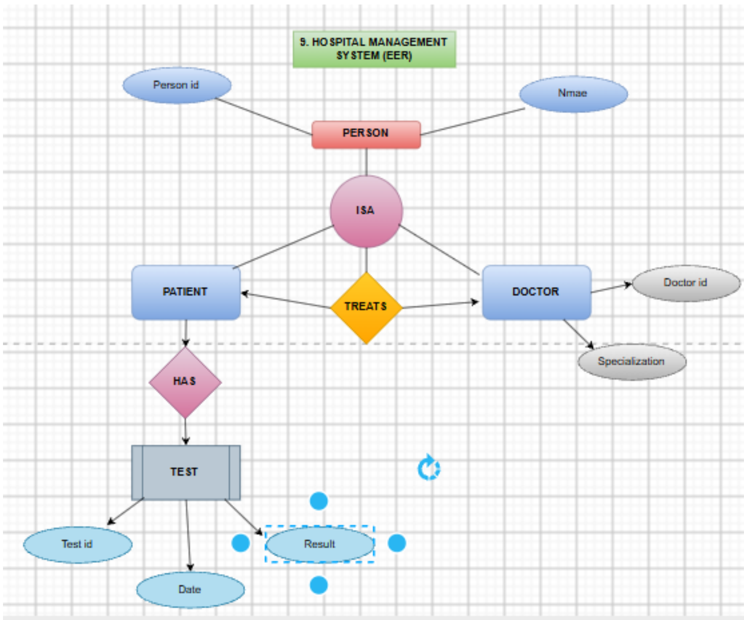
Physical Schema

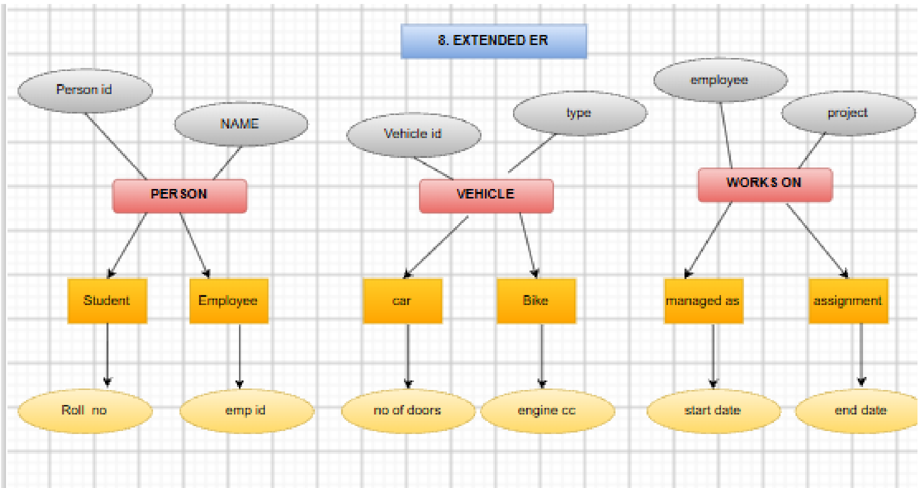
- Describes how data is stored on disk.
- Includes files, indexes, and storage methods.

View Schema

- Defines what data different users can see.
- Provides security by hiding unnecessary data.


7. BANK MANAGEMENT SYSTEM
Name
Customer id
Address
CUSTOMER
OWNS
TAKES
Account no
Blance
ACCOUNT
LOAN
Loan no
Ammount
Maintains
OPERATORS
Branch id
BRANCH
Branch name
Location


9. HOSPITAL MANAGEMENT SYSTEM (EER)
Person id
Nmae
PERSON
ISA
PATIENT
TREATS
DOCTOR
Doctor id
Specialization
HAS
TEST
Test id
Result
Date


8. EXTENDED ER
Person id
NAME
PERSON
Student
Employee
Roll no
emp id
Vehicle id
type
VEHICLE
car
Bike
no of doors
engine cc
employee
project
WORKS ON
managed as
assignment
start date
end date

10. Role of DBA (Database Administrator) (Short Answer)

Definition

A **Database Administrator (DBA)** is responsible for managing, securing, maintaining, and controlling the database to ensure efficient operation.

Responsibilities

- Install and configure the DBMS.
- Create and manage databases.
- Manage user accounts and permissions.
- Perform backup and recovery.
- Ensure database security.
- Monitor and improve performance.
- Maintain data integrity and consistency.
- Allocate storage and maintain database files.

Example:
In a college database, the DBA creates student accounts, grants faculty access to update marks, performs regular backups, and recovers data after system failures.